Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

Exercise 5.1 – A table

A survey covers 100 users of two transport services A and B. Of these, 40 use A, 35 use B and 15 use both. One of the 100 users is chosen at random; all are equally likely to be chosen. The events A and B mean, respectively, "the user uses service A" and "the user uses service B".

1. Build and complete a two-way table: use or non-use of A in the rows, use or non-use of B in the columns. Add the totals.
2. Calculate $P_A(B)$ and $P_B(A)$. For each calculation, state the reference population used as the denominator.
3. Interpret the two probabilities in two distinct sentences.

Exercise 5.2 – Production

A company makes objects in two workshops A and B, which provide $\frac{1}{4}$ and $\frac{3}{4}$ of the production respectively. Among the objects made by A, 4 percent are defective; among those made by B, 2 percent are. An object is chosen at random from the whole production. Let D be the event "the object is defective".

1. Represent the situation with a weighted tree and state the meaning of each second-level probability.
2. Calculate $P(D)$ and interpret the result as an overall defect rate.
3. It is now known that the chosen object is defective. What is the probability that it was made in workshop A?

Exercise 5.3 – Checking independence

A fair six-sided die, with faces numbered 1 to 6, is rolled. Consider the events A: "obtain an even number" and B: "obtain a number greater than or equal to 4".

1. Write the sets of outcomes corresponding to A, B and $A \cap B$.
2. Determine whether the events A and B are independent. Justify using a numerical criterion.
3. Explain why they are not disjoint either.

Exercise 5.4 – Without replacement

An urn contains three red balls and two blue balls, indistinguishable by touch. A first ball is drawn at random, then a second without replacing the first. At each draw, all the balls in the urn are equally likely.

1. Build the weighted tree of the two draws, distinguishing the colour obtained at the first and at the second draw.
2. Calculate the probability of drawing two red balls.
3. Calculate the probability of drawing one ball of each colour. Explain why two paths of the tree must be taken into account.

Exercise 5.5 – Three trials

An experiment has a success of probability $\dfrac{1}{3}$ and a failure. It is repeated three times independently, under the same conditions.

1. Build a three-level tree, or list the eight possible sequences of outcomes.
2. Calculate the probability of exactly one success. State the paths used.
3. Calculate the probability of at least two successes, treating the cases of two and of three successes separately.

Exercise 5.6 – Detector

A device checks objects of which 2 percent are defective. It raises an alarm for 90 percent of the defective objects and, wrongly, for 5 percent of the objects without a defect. An inspected object is chosen at random. Let D be the event “the object is defective” and S the event “the device raises an alarm”.

1. Determine the proportion of truly defective objects among those flagged. Choose a representation and detail the reasoning.
2. Give an exact value, then a percentage rounded to the nearest tenth.
3. Someone proposes 90 percent directly as the answer. Explain precisely which confusion of probabilities they are making.

Exercise 5.7 – Sets and cardinality

We work in the set $E = \{1; 2; 3; 4\}$. Consider its subsets $A = \{1; 2\}$ and $B = \{2; 4\}$.

1. Write $A \cap B$, $A \cup B$ and the complement of A in E. Give the cardinality of each.
2. Write all the pairs of $A \times B$, then give its cardinality.
3. Explain why the pairs $(1; 2)$ and $(2; 1)$ cannot be confused. Does the second belong to $A \times B$?

Exercise 5.8 – Delayed review

In a random experiment, two events A and B satisfy

$$P(A) = \frac{1}{2}, \qquad P_A(B) = \frac{1}{4}, \qquad P_{\overline{A}}(B) = \frac{3}{4}.$$

1. Calculate $P(B)$ using a tree or the law of total probability.
2. Calculate $P_B(A)$, making the common event used in the numerator explicit.
3. Are the probabilities $P_A(B)$ and $P_B(A)$ equal here? Is this equality a general property?

Homework. Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 67 p. 336, questions 1, 2 and 3. Build a tree keeping the textbook's events D and N. Careful: N denotes a negative test.

CQFD, exercise 70 p. 336, questions a and b. Count the four equally likely outcomes of the two throws to refute the proposed equalities.